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Tight Sand Reservoir Porosity Prediction Using Post-Stack Inversion

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In recent years, unconventional plays have emerged as a vital component of the global energy landscape, attracting significant economic attention due to their substantial potential for yielding valuable resources (Thakur, 2019), including a diverse range of oils (such as heavy oils, tight oils, and shale oils) and gas (including tight gas and shale gas) (Hongjun et al., 2016). The allure of these resources lies in their vast reserves and potential to meet the increasing demand for energy (Lin, 2016). However, despite the availability of these resources, they have been commercially underexploited in the past due to a multitude of reasons.

One of the primary challenges is the complex nature of the geology involved, which often features unconventional trapping methods that are driven by lithology rather than structural factors (Song et al., 2015). This complexity is further compounded by the fact that the field of unconventional exploration is relatively new, and as such, the geological, lithological, and technological advancements are still in the process of being developed and refined (Holditch, 2013). In contrast, conventional plays have been commercially exploited for a longer period, resulting in a more comprehensive understanding of the underlying geology and the development of more effective extraction methods (Campagna, 2012).

Another significant challenge is the expensive and challenging extraction methods involved, including stimulation efforts such as fracking and other enhanced oil recovery methods. These procedures require careful planning and execution, thus the planning stage prior to drilling becomes even more critical due to the associated economic risks. Consequently, the financial demands and technical complex nature of production can make it difficult to ensure the economic viability of unconventional play development (Bolonkin et al., 2014).

Background

In recent years, unconventional plays have emerged as a vital component of the global energy landscape, attracting significant economic attention due to their substantial potential for yielding valuable resources (Thakur, 2019), including a diverse range of oils (such as heavy oils, tight oils, and shale oils) and gas (including tight gas and shale gas) (Hongjun et al., 2016). The allure of these resources lies in their vast reserves and potential to meet the increasing demand for energy (Lin, 2016). However, despite the availability of these resources, they have been commercially underexploited in the past due to a multitude of reasons.

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However, the challenges associated with unconventional plays extend far beyond the operational aspects of drilling as they also impact earlier stages of the development process, ranging from seismic acquisition and processing to reservoir characterization and well placement. (Kaura et al., 2025). From a geophysics perspective, which is the primary focus of this study, one of the major challenges is acquiring sufficient seismic data that accurately maps deeper reservoirs. The low impedance contrast in shale plays often results in unresolvable seismic events, further complicating the process. Moreover, in developed conventional plays, there is often an abundance of shallow wells, but a scarcity of wells that penetrate to greater depths, leading to data sparsity that makes it difficult to correlate deeper seismic data with actual well measurements. These geophysical limitations introduce significant uncertainty into the planning process, which in turn increases the risks associated with subsequent stages like field evaluation and reservoir modeling. Furthermore, these prospective uncertainties later extend to the development phase, resulting in less effective well placement and reservoir management that ultimately affects the overall success of the project. The cumulative effect of these challenges underscores the many complexities typically associated with the life cycle of unconventional plays. As a result, the industry is committed to developing innovative solutions that can overcome such challenges, including the development of new technologies, methodologies, and workflows that improve the accuracy and efficiency of unconventional play development (Wilson, 2013). Consequently, such innovations play a crucial role in reducing the operational costs and associated risks, which can in turn unlock the full potential of unconventional resources for a more sustainable energy future that meets the increasing demand for energy (Muther et al., 2022).

The scope of this paper focuses on the geophysical challenges encountered during the reservoir characterization and quantitative interpretation phase, particularly porosity estimation. Porosity estimation plays an important role in the characterization of petroleum reservoirs as it directly impacts the estimated volumetrics of the field. However, porosity prediction becomes challenging in case of low porosity or tight sand reservoirs as the propagating seismic waves become less sensitive to the changes in porosity due to high stiffness and reduced fluid sensitivity. Consequently, a novel approach is being followed to predict reservoir porosity by conditioning the seismic data, well-calibrating seismic impedance inversion, and rock physics analysis tailored to this purpose.

Objectives

The primary objective of this study is to predict the porosity of an unconventional tight sand reservoir using post-stack inversion. However, the inherent complexities associated with unconventional plays pose significant challenges, including sparse well distribution, considerable depth of the target reservoir, and limited seismic resolution. Moreover, in the specific play under investigation, the presence of an overlying channel system exerts a profound impact on the seismic signal of the target reservoir, inducing velocity

pull-ups that give rise to artificial structures and misidentified faults, thereby further complicating the interpretation process.

This study presents three distinct workflows that were implemented as part of this project:

1. Utilization of regional well data to generate a prognosis for the target reservoir, which is subsequently employed to project hypothetical well tops in wells that do not penetrate the target reservoir.
2. Application of local corrections to the seismic interpretation that removes velocity pull-up structures caused by the overlying channel system, thereby enhancing the accuracy of the interpretation.
3. Implementation of post-stack inversion for porosity prediction, accompanied by a range of quality control (QC) measures to ensure the reliability and validity of the results.

To maintain confidentiality, the target reservoir will be referred to as Formation A, while the overlying formation will be designated as Formation B throughout this study. Additionally, no specific values or quantities will be addressed, ensuring that all discussions and presentations of results will be qualitative in nature, without disclosing any sensitive or proprietary information.

Methodology

Data Preparation

The seismic data utilized in this study consists of a reprocessed Pre-Stack Time Migrated (PSTM) 3D volume. Prior to conducting seismic interpretation, a comprehensive data collection and QC process was undertaken, encompassing all relevant data, including well coordinates, well logs, checkshots, and well tops. Subsequently, the area of interest's boundary was delineated by considering both the field's boundaries and the extent of well data coverage. Furthermore, key wells were identified for use in the post-stack inversion process, and a blind well was selected to be excluded from the full inversion and porosity prediction workflow, thereby facilitating a blind test of the predicted results.

Seismic Interpretation

The initial step preceding horizon mapping involved tying the wells to the seismic data, thereby establishing a direct correlation between the seismic data in the time domain and the actual well data in the depth domain. This was achieved by utilizing P-velocity logs (DTCO), density logs (RHOB), and available time-depth relationships (checkshots at either well location or from the nearest available well) to calculate the reflectivity and generate synthetic seismograms, which enable the seismic data to be tied to the existing well logs.

A total of three horizons, including the target reservoir, were identified and subsequently mapped manually (without snapping or extrapolation) across the area at 25×25 increment intervals. The interpretation process commenced with the selection of a cross-line that connected all the wells within the area of interest, and then the interpretation was extended to encompass the full extent of the area. The resulting interpretations were used to generate surfaces with a 50×50 grid increment, which would be utilized later for inversion purposes. To ensure the reliability of the results, the interpretation was cross-validated against similar interpretations in the same region, providing an additional layer of quality control and verification.

Formation Prognosis (Interpretation QC)

A supplementary QC step was developed which involves deriving the prognosis of the target Formation A. This was achieved by utilizing wells that penetrate the target formation from both the current play and nearby plays to generate a thickness map covering the *pre*-Formation A section. The well-derived thickness map was then regionally upscaled and calibrated (Fig. 2; a, b) to produce a regional thickness map that spans the area of interest, including at the shallow wells' locations. Subsequently, the estimated pre-Formation A

section was added to those shallow wells, resulting in pseudo well tops that represent Formation A (Fig. 2; c). These prognosed well tops were also subject to QC against deeper wells that include the actual tops of Formation A.

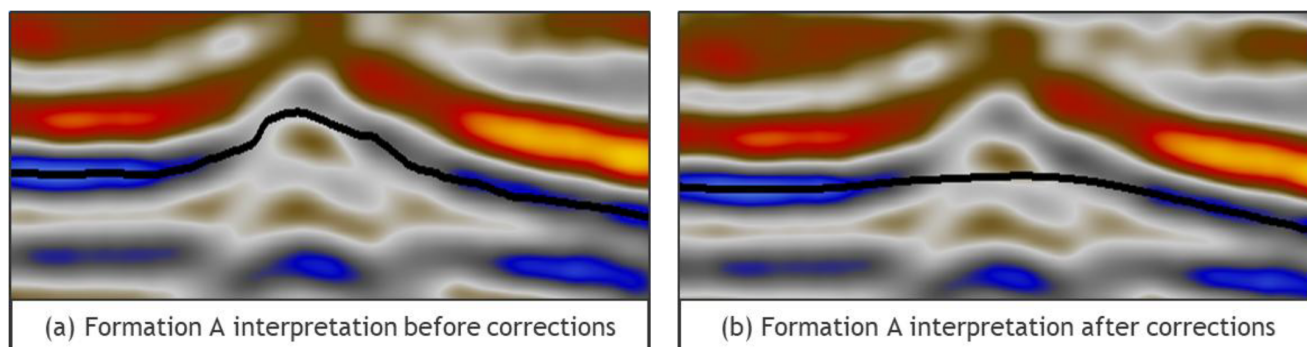


Figure 1—Seismic section showing an example of the local corrections applied to correct the velocity pull-ups.

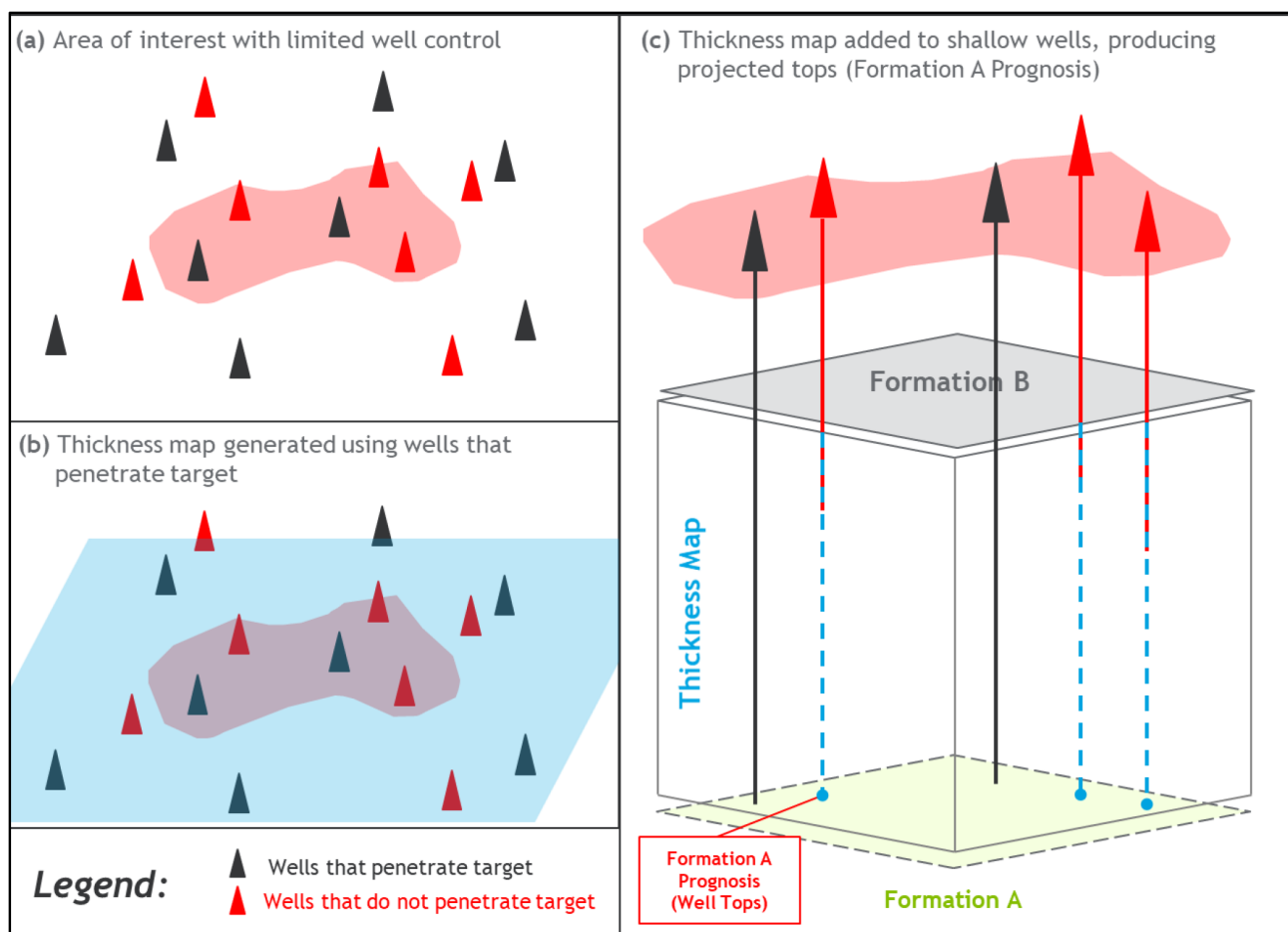


Figure 2—Schematic diagram illustrating the workflow used for Formation A prognosis: (a) Illustrating the well distribution (shallow versus deep) in the AOI. (b) Illustrating the regional thickness map generated using both wells from the AOI and nearby fields. (c) Illustrating how the thickness map was utilized to produce Formation A's prognosed pseudo-tops.

Channel Anomaly Correction

As stated previously, the area of interest is regionally affected by an overlying channel system, which introduces velocity pull-ups in the seismic data that could not be adequately resolved through re-processing alone. To address this issue, manual corrections were applied to the anomalies by first mapping the

boundaries of the overlying channels using seismic attributes as a guide, primarily relying on generalized spectral decomposition and variance. Generalized spectral decomposition is a method used to enhance the seismic resolution in regions where the seismic signal becomes unresolvable (e.g. thin beds below seismic resolution). As for variance, it is an edge detection method that measures the difference between neighboring seismic traces, often used to map seismic features related to faults, fractures, and channel boundaries, as well as other anomalies that cause a noticeable discontinuity in the seismic data.

Consequently, as the overlying channel system causes observable anomalies at the reservoir level, variance maps (flattened at the reservoir's level) were generated to highlight these channel-induced velocity pull-ups. Multiple variance maps were created at different frequencies to account for the range of apparent thicknesses of the anomalies, which are caused by variations in channel width and thickness. An RGB/CMY mixer was then used to enhance specific frequencies to maximize the signal of the channel-related anomalies, which was then used to regionally map the channels' extent.

Finally, local corrections were applied by estimating the pull-up time (ΔTWT) and subtracting it from the original interpretation, effectively eliminating the associated artificial structures (example shown in [Fig.1](#)).

Porosity Prediction (Post-stack Inversion)

For clastic reservoirs, the total porosity is a more representative parameter than the effective porosity as it accounts for both connected and unconnected pores, as well as the clay-bound water in a formation. Therefore, to establish the AI-porosity trend of our tight sand reservoir, the total porosity was chosen in favor of the effective porosity. Consequently, well log data was utilized to predict porosity via post-stack inversion by cross-plotting acoustic impedance (AI) on the x-axis against total porosity (PHIT) on the y-axis. In addition, Gamma Ray logs (GR), which measure the concentration of radioactive elements (Potassium, Uranium, and Thorium) within a sample were utilized in the z-axis as a pseudo-measure for lithology. For instance, clays are typically rich in radioactive minerals, thus a high GR reading is often correlated with elevated shale content, whereas lower GR readings are associated with less shale, suggesting a cleaner formation with higher sand content.

Several regression lines were tested, including linear, quadratic, and exponential models. Results indicated that a regression approach which divides the formation into two distinct linear trends yields the most accurate predictions. These two trends were differentiated based on the GR readings, resulting in the classification of a Sands trendline and a Shales trendline ([Fig.3](#)). Ultimately, as the focus of the study is on the tight sand section of the formation, the Sands trendline was selected to generate the porosity volume at the target depth.

As for the inversion workflow, the corrected surfaces and AI well logs were utilized to generate a low-frequency model (LFM), which accounts for the missing low frequencies in the seismic data. Subsequently, statistical wavelets with a Papoulis taper were extracted from each well, and these wavelets were then averaged to create a single *average multi-wavelet* wavelet. The LFM and the average wavelet were then used to invert the post-stack seismic data to obtain an inverted volume of acoustic impedance. In order to validate the inversion accuracy, the inverted impedance was carefully compared with the impedance from the blind well by cross-plotting the well measured impedance and the inverted impedance. Finally, the resulting AI volume was inputted into the previously established porosity equation, specifically the Sands trend, which enabled the population of porosity across the seismic-derived AI volume, thereby generating the 3D porosity model.

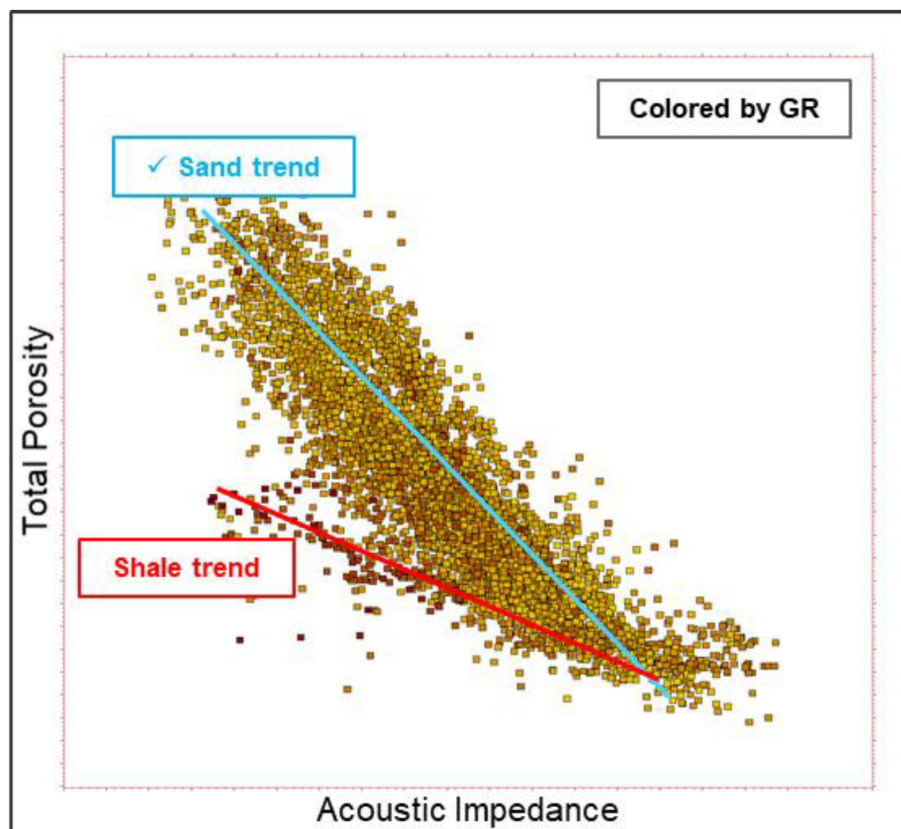


Figure 3—Total porosity versus acoustic impedance cross-plot (colored by Gamma Ray) showing the identified Sands and Shales trend.

Furthermore, in order to ascertain the accuracy of the prediction, a blind well was intentionally excluded from the full inversion and porosity prediction process. This blind well was then used to extract a seismic-derived porosity log, which was subsequently compared to the actual porosity log (filtered to match the seismic resolution) at that well location. Furthermore, the inversion workflow was repeated with several additional blind well tests to validate the porosity volume and increase confidence in the prediction, providing an independent validation of the results and ensuring the accuracy and reliability of the porosity model.

Results & Discussion

The relatively recent interest in unconventional plays gives rise to a multitude of challenges that affects not only the field development phase but also the earlier, more technical stages such as field appraisal and reservoir characterization and evaluation. One of the common challenges includes the scarcity of data available to understand the deeper subsurface. For instance, the unavailability of deeper well logs that penetrate the target prevents adequate quantitative estimates of formation parameters, and also limits the ability to establish a reliable relationship between well logs and seismic data. Moreover, legacy seismic data is acquired with shallower targets in mind, which is often of poor resolvability at greater depths due to amplitude loss, as well as structural and lithologic factors such as interbedding, compaction, and deformation of the deeper subsurface. However, collecting new seismic data is costly and often not prioritized in previously explored regions, leading to the increased re-utilization of legacy data for unconventional exploration. Nevertheless, advanced re-processing techniques that focus on enhancing the deeper geology have made it possible to utilize previously acquired seismic data for unconventional reservoir characterization purposes without the added cost of re-acquisition. As a result, this study utilizes a *reprocessed* Pre-Stack Time Migrated 3D volume, which significantly enhances legacy data in terms of

event resolvability and signal-to-noise ratio at the reservoir's depth. This improvement addresses challenges associated with imaging at target depths, including high-frequency noise, seismic attenuation, and multiples that interfere with the target signal, allowing for a more accurate characterization of subsurface geology. The reprocessed data also improves the imaging quality of faults and structures previously considered uncertain, later identified as artificial structures caused by channel velocity pull-ups.

Beyond conventional QC methods, an additional QC step was implemented to further ensure the accuracy and reliability of the result, which involves deriving the prognosis of the target Formation A. This is necessary because most wells in the area do not penetrate the target reservoir, and this approach enables further constraining of the interpretation in areas where wells are present but do not reach the target formation. The prognosed Formation A well tops were tested against deeper wells that include the actual tops of Formation A, yielding satisfactory results that validate the approach. This method serves as an additional *well-based* QC method, allowing for the utilization of shallow wells that are more abundant than deeper wells in the area but were otherwise not useful. By leveraging these shallow wells, the approach enhances the overall quality and reliability of the interpretation, and provides a more comprehensive, well-based understanding of the subsurface geology.

An additional QC step was implemented which involved correcting seismic anomalies caused by the overlying channel system that regionally affects the area. This correction process successfully removed velocity pull-up-related anomalies, resulting in a more accurate representation of the subsurface geology. A comparison of the corrected interpretation with non-corrected interpretations of the same region revealed that previously questioned structures, such as structural highs and fault-like features, were successfully eliminated after correction. Moreover, some previously unquestioned features were also determined to actually be anomaly-related. As a result, the corrected interpretation yielded a refined regional model with more geologically plausible features that are consistent with the tectonic history of the region. Furthermore, the results were cross-validated against nearby fields with similar characteristics, specifically those exhibiting seismic anomalies due to overlying channel systems, and the findings remained consistent, thereby increasing confidence in the reliability of the applied corrections.

Finally, the inversion process yielded optimal results when the formation lithology was differentiated based on gamma ray readings, effectively distinguishing between sands and shales. This approach enabled the creation of a more representative porosity cube that accurately honors the tight sand reservoir of interest. However, it is noted that this porosity cube does not accurately represent the porosity of non-reservoir zones as the applied AI-Porosity transform equation excludes the non-reservoir zones. Furthermore, due to the inherent limitations of seismic resolution, the resulting porosity cube is more suited for identifying regional trends and providing qualitative insights, rather than quantitative predictions.

Conclusions

In conclusion, this workflow introduces a novel approach that utilizes channel boundaries to identify areas of potential false seismic features, thereby significantly enhancing the quality of the interpretation by removing previously questioned structures. Furthermore, this approach also revealed that some structures that were initially considered valid were, in fact, false, leading to a refined understanding and re-evaluation of the area. The validation of these results was further reinforced through cross-validation against nearby fields that are similarly affected by the overlying channel system, where comparable findings were observed, thus increasing confidence in the accuracy and reliability of the corrected interpretations.

The successful integration of the corrected interpretations into the inversion workflow to predict porosity demonstrates the practical application and value of this innovative approach. Moreover, the potential of this approach extends beyond porosity prediction, as it can also be used to build more accurate geomodels, which is a critical component of reservoir characterization and field development planning. Although the

construction of the geomodel is beyond the scope of this paper, the corrected interpretations provide a solid foundation for future geomodeling efforts.

The significance of this workflow lies in its ability to improve the accuracy and reliability of subsurface geological interpretations, which is essential for optimizing unconventional field development and production strategies. By leveraging channel boundaries to identify and remove false structures, this approach enables a more nuanced understanding of the subsurface geology, which can lead to better-informed decisions regarding well placement, reservoir management, and hydrocarbon recovery. Ultimately, the successful application of this innovative approach has the potential to reduce uncertainty, minimize risks, and maximize returns in the development of unconventional plays, making it a valuable tool for the oil and gas industry.

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